Low-Dose PET Image Enhancement Using CycleGAN

Abhishek Raman

Abstract

Positron emission tomography (PET) scans are an invaluable functional imaging modality in oncology and neurology, and they provide quantitative information about tissue metabolism and receptor density that anatomical techniques can't offer. High-quality PET acquisition requires a high radiotracer dose, typically 3–7 MBq kg^{-1}, resulting in effective radiation doses of 7–14 mSv per examination. Higher levels of radioactive tracer exposure can be dangerous for patients, but lower exposure degrades image quality and introduces noise that reduces the reliability of standardized uptake value (SUV) measurements. This project explores the use of CycleGAN, an unpaired image-to-image translation framework, can enhance low-dose PET scans to full-dose PET scans without requiring high radioactive tracers. Using the UDPET Ultra-Low Dose PET Imaging Challenge dataset of 1,447 whole-body ^{18}F-FDG scans, a preprocessing pipeline was eveloped to convert raw DICOM volumes into per-slice normalized, spatially standardized HDF5 archives. A CycleGAN with a ResNet-based generator and PatchGAN discriminator was trained to map low-dose coronal PET slices to the full-dose domain using cycle consistency and identity losses. The model was then evaluated on 15146 slices and showed consistent improvement across six metrics. Mean PSNR improved by $+5.97$ dB (from 29.71 to 35.68 dB), SSIM improved by $+0.124$ (from 0.439 to 0.563), and perceptual distance as measured by LPIPS decreased by 74.6%. Improvement was observed in all 25 test patients. However, 3.4% of slices experienced quality degradation relative to the low-dose input, with worst-case PSNR reductions of up to -17.1 dB. The results show promise for the use of deep-learning based methods in clinical settings; the model recovered clinically meaningful image quality from low-dose inputs across patients.

Keywords: PET imaging, CycleGAN, generative adversarial networks, medical image enhancement, image-to-image translation, deep learning, image quality metrics.

1. Introduction

1.1 Positron Emission Tomography: Physical Principles and Clinical Role

PET is a crucial modality that uses in vivo biological processes with spatial and temporal specificity that can't be achieved through structural techniques like computed tomography (CT) or magnetic resonance imaging (MRI). Rather than depicting macroscopic anatomy, PET maps the three-dimensional distribution of a positron-emitting radiotracer, thereby reflecting regional metabolic activity, receptor density, perfusion, or gene expression. This functional sensitivity renders PET indispensable in fields where tissue physiology governs diagnosis, prognosis, and therapeutic decision-making.

The underlying mechanism of PET depends on the kinematics of positron-electron annihilation. A radionuclide that emits a positron, such as fluorine-18 (^{18}F) conjugated to the glucose analogue 2-deoxy-2-(^{18}F)fluoro-D-glucose (^{18}F-FDG), is administered intravenously and accumulates in tissues with elevated glycolytic demand. These tissues are often the sight of tumor cells that upregulate glucose transporters (GLUT-1, GLUT-3) and hexokinase.

The main quantitative measure used in PET imaging is the standardized uptake value (SUV), defined as:

$$\mathrm{SUV} = \frac{C(t)}{D/BW}, \tag{1}$$

where $C(t)$ is the tracer concentration in tissue at time t, D is the injected dose, and BW is the patient's body weight. SUV helps compare how much tracer different tissues absorb and is commonly used in clinical settings, for example when evaluating how a tumor is responding to treatment. However, SUV measurements depend heavily on image quality. When PET scans are acquired at low dose, the reduced photon counts introduce more noise, which can make SUV estimates less reliable and potentially affect clinical decisions.

1.2 Low-Dose PET: The Noise-Dose Tradeoff

Standard ^{18}F-FDG whole-body PET requires approximately 3–7 MBq kg^{-1} injected activity that correspondes to an effective dose of roughly 7–14 mSv. Reducing the administered activity proportionally reduces the expected coincidence count rate. For Poisson-distributed counts N, the signal-to-noise ratio scales as:

$$\mathrm{SNR} \propto \sqrt{N}, \tag{2}$$

so halving the dose reduces SNR by a factor of $\sqrt{2} \approx 1.41$. This leads to more visible noise in the reconstructed image, lower contrast in small lesions, and reduced visibility of regions with weak tracer uptake. When the count level becomes very low, small lesions may not be detected at all. In addition, SUV estimates become less stable and may vary enough to affect how treatment response is interpreted.

Even with these limitations, reducing dose is still important in many clinical settings. Children are more sensitive to radiation and have a longer lifetime over which radiation effects may appear, so lower dose protocols are often preferred. Patients who require multiple PET scans over time, such as those undergoing treatment monitoring, can also accumulate significant total radiation exposure. New scanner designs with longer axial coverage improve sensitivity, but they are not widely available. Because of this, there is strong interest in computational methods that can improve image quality from low-count data without increasing the administered dose.

1.3 CycleGAN for Unpaired Image Translation

CycleGAN [3] extends adversarial image translation to the unpaired setting. Two generators, $G_{AB} : A \to B$ and $G_{BA} : B \to A$, along with two discriminators D_A and D_B, are trained simultaneously. D_B distinguishes real samples $y \in B$ from translated outputs $G_{AB}(x)$, while D_A distinguishes real samples $x \in A$ from back-translated outputs $G_{BA}(y)$. The adversarial losses alone are insufficient to prevent mode collapse, since infinitely many mappings can satisfy them without actually preserving the content of the input image. CycleGAN addresses this problem through a *cycle consistency loss*:

$$\mathscr{L}_{\text{cyc}} = \mathbb{E}_{x\sim p_A}\left[\|G_{BA}(G_{AB}(x)) - x\|_1\right] + \mathbb{E}_{y\sim p_B}\left[\|G_{AB}(G_{BA}(y)) - y\|_1\right], \tag{3}$$

which forces the two generators to behave as approximate inverses of one another, ensuring that a source image can be recovered after a round-trip translation through both domains. The full training objective combines the adversarial and consistency terms:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{adv}} + \lambda\,\mathscr{L}_{\text{cyc}}, \tag{4}$$

where $\lambda = 10$ in the original formulation. An optional *identity loss*, $\mathscr{L}_{\text{idt}} = \|G_{AB}(y) - y\|_1 + \|G_{BA}(x) - x\|_1$, penalizes unnecessary modification of images that already belong to the target domain. This term is particularly relevant in PET imaging because SUV magnitudes carry direct quantitative biological meaning, and any gratuitous rescaling of activity values could compromise the clinical utility of the enhanced output.

Generator architecture. As shown in Figure 1, each generator follows a ResNet-based encoder-decoder design. The encoder applies two strided ($s = 2$) convolutional layers that reduce the spatial resolution by a factor of four while expanding the channel depth to 256. The resulting feature representation then passes through a bank of 6 or 9 residual blocks, each consisting of two 3×3 convolutional layers with instance normalization and ReLU activation connected by a skip connection. These blocks apply nonlinear transformation without changing the spatial dimensions. The decoder mirrors the encoder structure, using transposed convolutions to progressively restore the original resolution. Instance normalization is used throughout the network rather than batch normalization, since batch normalization depends on statistics

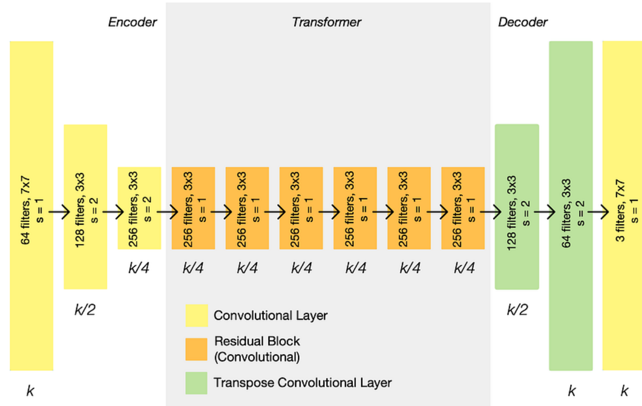

Figure 1. CycleGAN generator (ResNet-based encoder-decoder). Two strided convolutional layers ($s = 2$) compress the input from $k \times k$ to $k/4 \times k/4$ at 256 channels. A series of residual blocks apply nonlinear transformation in this compressed space. Two transposed convolutional layers restore the original spatial resolution.

computed across the mini-batch and was found empirically to perform worse for image translation tasks where the effective batch size is small.

Discriminator architecture. The PatchGAN discriminator shown in Figure 2 is composed of four convolutional blocks, each using 4×4 filters with stride $s = 2$, interleaved with instance normalization and Leaky ReLU activation (negative slope $\alpha = 0.2$). Each strided block halves the spatial resolution, taking the representation from $k/2$ through $k/4$ and $k/8$ down to $k/16$. A final 4×4 convolutional layer followed by sigmoid activation produces a single-channel output map in which each scalar value corresponds to a local patch of the input image. For the four-layer configuration described here, the receptive field of each output scalar spans approximately 70×70 pixels. This patch-level evaluation makes the discriminator insensitive to global image composition and concentrates its capacity on local texture, which is both where generative models most frequently introduce artifacts and where fine structural detail relevant to medical interpretation tends to reside.

1.4 Prior Work in Low-Dose PET Enhancement

Deep learning approaches to PET denoising and enhancement have expanded substantially since Chen et al. [4] showed that a residual encoder-decoder network trained on count-decimated data could recover a meaningful portion of the image quality lost at quarter-dose acquisition. Subsequent studies explored U-Net variants, dense convolutional networks (DenseNets), and attention-gated encoder-decoder architectures, often pairing pixel-level L1 or L2 reconstruction losses with perceptual losses computed in the feature space of a pretrained VGG network. The perceptual terms encourage sharper edges and better-preserved fine texture compared to regression objectives alone.

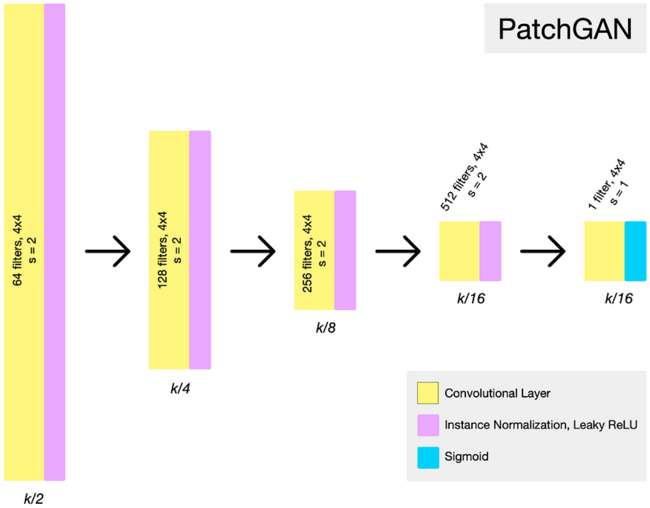

Figure 2. CycleGAN PatchGAN discriminator. Four successive 4×4 convolutional blocks ($s = 2$) with instance normalization and Leaky ReLU progressively halve the spatial resolution from $k/2$ to $k/16$. The final layer produces a patch-level real/fake probability map via sigmoid activation.

GAN-based methods were introduced specifically to address the tendency of purely supervised regressors to over-smooth fine structural detail. Lei et al. [5] trained a conditional GAN on paired low- and full-dose brain PET data and reported improved PSNR and SSIM over OSEM reconstruction. Luo et al. [6] applied a similar strategy to whole-body FDG-PET and found that SUV measurements in the enhanced images remained within 5% of the full-dose reference across a heterogeneous cohort of lesion types, suggesting that adversarial training can be made compatible with quantitative accuracy requirements under controlled conditions.

That said, GAN-based enhancement carries a well-documented risk of *hallucination*, whereby the generator synthesizes plausible-looking structure that has no correspondence to the true underlying activity distribution. In PET, this failure mode could take the form of spurious uptake foci that mimic genuine lesions, suppression of real but faint tracer accumulation through spatial averaging toward the background, or distortion of lesion boundary morphology in ways that alter the apparent metabolic tumor volume. What makes this particularly problematic from a validation standpoint is that hallucinated outputs can score well on pixel-level metrics such as PSNR and SSIM relative to the reference image, since those metrics reward global similarity and are not designed to detect localized structural fabrication. This gap between standard image quality measures and clinical fidelity is a recurring theme throughout this paper.

1.5 Image Quality Metrics and Their Clinical Relevance

Quantitative evaluation of PET enhancement algorithms relies on a combination of pixel level fidelity measures, perceptual quality indices, and clinically motivated quantities. Six metrics were used in this work, each capturing a distinct aspect of image quality. A shared limitation across all of them is that none directly measures the clinical correctness of specific structures, a point taken up in Section **??**.

Peak signal to noise ratio (PSNR) measures the ratio of the maximum possible signal power to the mean squared reconstruction error:

$$\mathrm{PSNR} = 10 \log_{10}\left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}}\right), \quad \mathrm{MSE} = \frac{1}{N}\sum_i (\hat{x}_i - x_i)^2, \quad (5)$$

where MAX is the maximum possible pixel intensity, $\hat{x}_i$ and x_i are corresponding enhanced and reference pixel values, and N is the total pixel count. PSNR is expressed in decibels and values in the range 25 to 38 dB are typical for PET enhancement tasks. Its central weakness is that it treats all pixels equally regardless of spatial context, so two images with identical MSE but different spatial error distributions score identically even if one contains a lesion shaped artifact.

Normalized mean squared error (NMSE) scales the pixel level squared error by the energy of the reference image:

$$\mathrm{NMSE} = \frac{\sum_i (\hat{x}_i - x_i)^2}{\sum_i x_i^2}. \quad (6)$$

Unlike raw MSE, NMSE is dimensionless and invariant to the global intensity scale of the reference, making it more comparable across patients with different administered activities. A value of zero indicates perfect reconstruction while values approaching one indicate reconstruction error on the order of the signal energy itself. NMSE shares PSNR's insensitivity to spatial structure and can be driven by errors in homogeneous background regions rather than diagnostically relevant uptake regions.

Structural similarity index measure (SSIM) [7] addresses the spatial blindness of pixel error metrics by decomposing image fidelity into three components, luminance l, contrast c, and structure s, evaluated within a sliding local window:

$$\mathrm{SSIM}(\hat{x}, x) = l(\hat{x}, x) \cdot c(\hat{x}, x) \cdot s(\hat{x}, x) \ \in [-1, 1], \quad (7)$$

where each term is a ratio of the corresponding local statistics such as means, standard deviations, and cross correlation stabilized by small constants C_1 and C_2. The mean SSIM over all windows is reported as MSSIM and ranges from -1 to 1, with 1 denoting a perfect match. SSIM correlates better with human perceptual judgments than PSNR because edge blurring and texture distortions reduce the structure term even when pixel level errors remain small. However, window based evaluation means that globally inconsistent intensity rescaling can go undetected if local statistics are preserved.

Learned perceptual image patch similarity (LPIPS) [10] moves beyond handcrafted statistics by measuring the distance between enhanced and reference images in the feature space of a deep network pretrained on a large visual dataset:

$$\text{LPIPS}(\hat{x}, x) = \sum_{\ell} w_\ell \, \|\phi_\ell(\hat{x}) - \phi_\ell(x)\|_2^2, \tag{8}$$

where ϕ_ℓ denotes the feature map at layer ℓ of the pretrained network and w_ℓ is a learned weighting. Lower LPIPS indicates greater perceptual similarity. Because the features encode semantic and textural structure learned from natural images, LPIPS is sensitive to distortions such as blurring, artificial texture, and contrast inversion. Its limitation in the medical context is that the pretrained features reflect natural image statistics rather than the structural properties specific to PET, so it may not weight clinically meaningful regions appropriately.

Gradient magnitude similarity deviation (GMSD) [11] evaluates image quality by comparing pixel wise gradient magnitudes between the enhanced and reference images:

$$\text{GMSD}(\hat{x}, x) = \sqrt{\frac{1}{N} \sum_{i} \left(\text{GMS}_i(\hat{x}, x) - \overline{\text{GMS}}\right)^2}, \tag{9}$$

where GMS_i is the local gradient magnitude similarity at pixel i and $\overline{\text{GMS}}$ is its image wide mean. Lower GMSD indicates higher quality. The metric is sensitive to edges and fine structures, which are important in PET because lesion boundaries determine both visual detection and quantitative measurements such as tumor volume.

Visual information fidelity in the pixel domain (VIFp) [12] takes an information theoretic approach, quantifying the mutual information between the reference image and the enhanced image as seen through a model of the human visual system and an additive noise channel:

$$\text{VIFp} = \frac{\sum_{\mathbf{k} \in \text{subbands}} I(\mathbf{C}^{\mathbf{k}}; \mathbf{F}^{\mathbf{k}} | s_s)}{\sum_{\mathbf{k} \in \text{subbands}} I(\mathbf{C}^{\mathbf{k}}; \mathbf{E}^{\mathbf{k}} | s_s)}, \tag{10}$$

where the numerator captures the visual information that can be extracted from the enhanced image and the denominator captures the information available in the original reference. Higher VIFp indicates greater fidelity. In practice, VIFp is particularly sensitive to blurring, which is a common artifact in PET enhancement models.

Table 1 summarizes all six metrics together with their definitions, directional convention, and typical operating ranges.

1.6 Scope and Contributions of This Work

This project explores whether CycleGAN can enhance low-dose PET images by translating them into images that resemble full-dose PET without requiring paired training data. The main idea is that the cycle consistency constraint can help preserve important structural and quantitative information, such as SUV, during this translation. PET DICOM data from the UDPET dataset are preprocessed using per-slice normalization, spatial standardization, and HDF5 storage before training the model.

Table 1. Image quality metrics used to evaluate PET enhancement. Arrows indicate the direction of improvement.

Metric	Definition or key property	Range
PSNR↑	$10\log_{10}(\text{MAX}^2/\text{MSE})$; pixel level	25 to 38 dB
NMSE↓	Normalized squared error; scale invariant	≥ 0
SSIM↑	Luminance times contrast times structure	$[-1, 1]$
LPIPS↓	Deep feature perceptual distance	[0, 1]
GMSD↓	Gradient magnitude deviation	≥ 0
VIFp↑	Information theoretic fidelity	≥ 0

Note: No metric directly measures lesion level diagnostic correctness. High PSNR or SSIM does not exclude hallucinations, and LPIPS reflects natural image features rather than PET specific structure.

The results are evaluated by comparing enhanced images with low-dose inputs and full-dose references using both visual inspection and standard image quality metrics. This is a high-risk problem because improving visual quality does not guarantee clinical accuracy. In PET imaging, even small distortions can affect lesion detection or treatment assessment. Therefore, the goal of this work is not only to test whether CycleGAN improves image quality, but also to understand its limitations, including cases where it may introduce misleading structures or alter important quantitative features.

2. Methodology

2.1 Dataset

For thsi project, training was done using the UDPET (Ultra-Low Dose PET Imaging Challenge) dataset. This dataset consists of 1,447 whole-body ^{18}F-PET scans acquired on a uExplorer total-body PET/CT scanner. Each patient record has eight dose variants that are stored in separate DICOM formats: one full-dose reference (denoted `NORMAL`) and seven progressively reduced acquisitions at 2×, 4×, 5×, 10×, 20×, 50×, and 100× dose reduction factors. The lower dose images are acquired by sinogram decimation to ensure that each dose pair is spatially registered by construction.

2.2 Preprocessing Pipeline

2.2.1 DICOM Loading

All `.dcm` files within each dose-specific folder were read in sorted order using `pydicom`, converted to `float32` arrays, and stacked into 3D volumes of shape (N_s, H, W). Here, $N_s \approx 600$–700 denotes the number of axial slicesm and the native in-plane resolution is 360×360 pixels. I also discarded volumes with fewer than one valid slice.

2.2.2 Per-Slice Intensity Normalization

Intensity normalization was applied independently to each 2D axial slice rather than globally across the volume to account for the large slice-to-slice variation in radiotracer uptake inherent to whole-body PET. Each slice was rescaled to $[-1,1]$ via min-max normalization:

$$\tilde{s} = \frac{s - s_{\min}}{s_{\max} - s_{\min}} \cdot 2 - 1, \tag{11}$$

where $s_{\min}$ and $s_{\max}$ are the per-slice intensity extrema. Slices with $s_{\max} = s_{\min}$ (blank slices) were set to zero. This range is compatible with the generator's `Tanh` output activation and avoids the SUV-dynamic-range collapse that global normalization would introduce.

2.2.3 *Spatial Standardization*

All slices were standardized to a fixed spatial resolution of 640×320 pixels (height $\times$ width). For slices larger than the target along a given axis, a center crop was applied:

$$\text{start} = \lfloor (d_{\text{in}} - d_{\text{target}})/2 \rfloor, \tag{12}$$

extracting the central d_{target} pixels. For slices smaller than the target, symmetric zero-padding was applied:

$$p_{\text{before}} = \lfloor (d_{\text{target}} - d_{\text{in}})/2 \rfloor, \quad p_{\text{after}} = d_{\text{target}} - d_{\text{in}} - p_{\text{before}}. \tag{13}$$

The original 360×360 DICOM slices are center-padded along the height axis and center-cropped along the width axis to reach 640×320.

2.2.4 *HDF5 Storage*

Preprocessed volumes for all eight dose keys were stored in a single `.h5` file per patient, organized as:

```
Patient_ID.h5
  full     (N, 640, 320)  -- full-dose reference
  1_2      (N, 640, 320)  -- 2x reduction
  ...
  1_100    (N, 640, 320)  -- 100x reduction
```

Datasets were stored as `float32` with `gzip` compression at level 4, reducing per-patient storage from ∼200 MB (raw DICOM) to ∼5–15 MB. This layout allows flexible key-based selection of any input/target dose pair at training time without reloading the full volume.

2.3 Model Architecture

2.3.1 *Generator: U-Net*

The generator follows the U-Net architecture [8] with skip connections between corresponding encoder and decoder layers. The input is a single-channel ($C = 1$) grayscale PET slice, and the output is a single-channel enhanced slice. The encoder applies successive downsampling blocks consisting of strided convolution, batch or instance normalization, and ReLU activation. The decoder then mirrors the encoder with transposed convolutions, concatenating skip-connected encoder feature maps at each resolution level to preserve spatial detail. The final layer uses `Tanh` activation to produce outputs in $[-1,1]$, consistent with the normalization in Equation (11).

2.3.2 *Discriminator: PatchGAN*

The discriminator follows the PatchGAN design operating on concatenated input–output imagepairs. Instance normalization and Leaky ReLU (slope $\alpha = 0.2$) are used throughout. The 70×70 receptive field focuses discrimination on local texture statistics rather than global image composition.

2.4 Training Procedure

2.4.1 *Multi-Key Orchestration*

A separate model was trained for each dose reduction factor $k \in \{2, 5, 10, 20, 50, 100\}$. For each factor, the corresponding low-dose key (`1_k`) was used as the generator input and the full-dose key (`full`) as the reconstruction target. This strategy allows each model to learn the noise characteristics specific to its dose level..

2.4.2 *Loss Functions*

CycleGAN (unpaired). For the unpaired CycleGAN variant, the objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{adv}} + \lambda\, \mathcal{L}_{\text{cyc}} + \mathcal{L}_{\text{idt}}, \tag{14}$$

Here, $\mathcal{L}_{\text{cyc}}$ is the cycle consistency loss defined in Equation (3) and $\mathcal{L}_{\text{idt}}$ is the identity loss that penalizes unnecessary modification of images already belonging to the target domain.

2.4.3 *Optimization*

Both models were trained with the Adam optimizer. A summary of all hyperparameters is given in Table 2.

The learning rate was held fixed at 2×10^{-4} for the first 100 epochs and then linearly annealed to zero over the subsequent 100 epochs, following the schedule of Zhu et al. [3]. A discriminator history buffer of size 50 was used to smooth discriminator updates by sampling from previously generated images [9].

2.5 Evaluation

Enhanced images produced by the trained generator were evaluated against the full-dose reference using the six image metrics that were computed slice-by-slice and averaged over the test set. Low-dose inputs were also scored against the full-dose reference to provide a baseline; this allowed for an assessment of the net improvement attributable to the enhancement model.

Table 2. Training hyperparameters.

Parameter	**Description**	**Value**
Optimizer		
lr	Initial learning rate (Adam)	0.0002
β_1	Adam momentum term	0.5
lr policy	Schedule type	linear
Schedule		
n_{ep}	Epochs at fixed lr	100
n_{decay}	Epochs with linear decay	100
Total epochs	Full training duration	200
Batch size	Images per update	1
Loss		
GAN mode	Adversarial loss variant	LSGAN
λ	L1 / cycle loss weight	100
Pool size	Discriminator history buffer	50
Architecture		
Norm type	Normalization layer	instance
Init type	Weight initialization	normal
Init gain	Init scale factor	0.02

3. Results

3.1 Overview of Evaluation Dataset

The trained CycleGAN was evaluated on a held-out test set comprising 25 patients and 15,146 coronal PET slices (60–135 slices per patient, median 98). For each slice, six metrics were computed between the enhanced output and the full-dose reference: PSNR, SSIM, NMSE, LPIPS, GMSD, and VIFp. The same metrics were computed for the raw low-dose input to establish a no-enhancement baseline. Reported improvements are defined as $\Delta m = m_{\text{enhanced}} - m_{\text{input}}$ for metrics where higher is better (PSNR, SSIM, VIFp), and $\Delta m = m_{\text{input}} - m_{\text{enhanced}}$ for error metrics where lower is better (NMSE, LPIPS, GMSD).

3.2 Aggregate Metric Performance

Table 3 reports mean, median, and standard deviation of all six metrics across all 15,146 slices. Figure 3 visualizes the aggregate comparison between input and enhanced images.

The CycleGAN enhanced images outperformed the low-dose input on all six metrics. PSNR improved by a mean of +5.97 dB (from 29.71 to 35.68 dB), and SSIM improved by +0.124 (from 0.439 to 0.563). Perceptual quality improved substantially, with LPIPS reducing from 0.280 to 0.071 — a 74.6% reduction in perceptual distance from the full-dose reference. The VIFp score more than doubled from 0.160 to 0.375. NMSE decreased from 0.413 to 0.136, though its high standard deviation (±0.424) indicates substantial per-slice variance in reconstruction quality.

3.3 Per-Patient Results

Figure 4 shows per-patient PSNR and SSIM for all 25 test subjects, and Table 4 provides the complete numerical summary. Improvements were consistent across patients since every patient showed a positive mean PSNR improvement, ranging from +3.21 dB (P20) to +7.77 dB (P24), and every patient showed a positive mean SSIM improvement, ranging from +0.070 (P04) to +0.237 (P11).

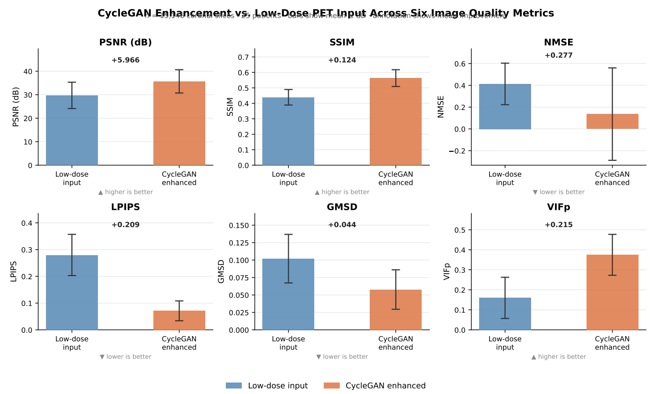

Figure 3. Aggregate metric comparison between low-dose input and CycleGAN enhanced images across all 15,146 test slices (25 patients). For NMSE, LPIPS, and GMSD, lower values are better; for all others, higher values are better.

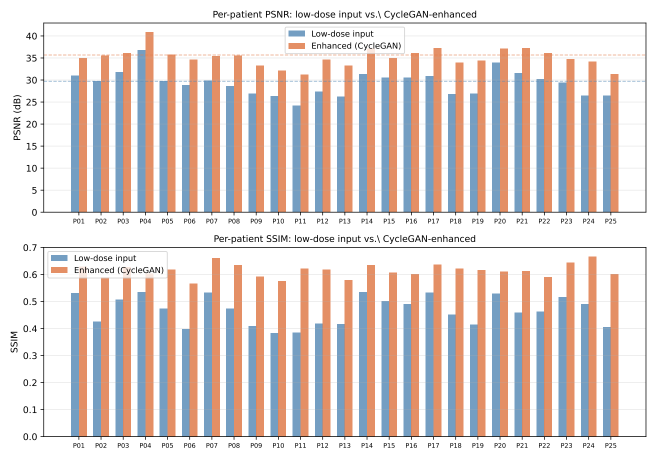

Figure 4. Per-patient PSNR (top) and SSIM (bottom) for low-dose input (blue) and CycleGAN-enhanced output (orange) across all 25 test patients (P01–P25). Dashed horizontal lines indicate the dataset-wide means. Enhancement is consistent across all patients.

3.4 Improvement Distributions and Failure Modes

Figure 5 shows the distribution of per-slice PSNR, SSIM, and LPIPS improvements across all 15,146 slices.

The improvement distributions for PSNR and SSIM are right skewed, with median gains of +6.52 dB and +0.121 respectively. Both medians are higher than the corresponding means, which suggests that most slices benefit from the enhancement. However, 521 out of 15,146 slices (3.4%) showed a negative change in PSNR, meaning the CycleGAN output performed worse than the low dose input for those

Table 3. Aggregate image quality metrics across 15,146 test slices (25 patients).

Metric	Low-dose input		CycleGAN enhanced		Improvement	
	Mean	Median	Mean	Median	Mean ± SD	Median
PSNR (dB)↑	29.71	28.81	35.68	35.23	+5.97 ± 2.73	+6.52
SSIM↑	0.439	0.443	0.563	0.569	+0.124 ± 0.060	+0.121
NMSE↓	0.413	0.389	0.136	0.081	−0.277 ± 0.396	−0.286
LPIPS↓	0.280	0.289	0.071	0.068	−0.209 ± 0.060	−0.215
GMSD↓	0.102	0.104	0.058	0.056	−0.044 ± 0.020	−0.043
VIFp↑	0.160	0.134	0.375	0.366	+0.215 ± 0.069	+0.222

Table 4. Per-patient mean PSNR and SSIM for low-dose input and CycleGAN enhanced output.

		PSNR (dB)			SSIM		
ID	*N*	In	Enh	Δ	In	Enh	Δ
P01	66	30.96	34.97	+4.01	0.530	0.629	+0.099
P02	93	29.82	35.60	+5.78	0.426	0.589	+0.163
P03	98	31.85	36.07	+4.22	0.508	0.627	+0.119
P04	60	36.76	40.89	+4.13	0.534	0.604	+0.070
P05	108	29.74	35.73	+6.00	0.473	0.619	+0.145
P06	106	28.83	34.69	+5.86	0.397	0.567	+0.170
P07	82	29.89	35.48	+5.59	0.532	0.661	+0.129
P08	116	28.68	35.55	+6.88	0.474	0.636	+0.162
P09	116	26.88	33.29	+6.41	0.409	0.593	+0.184
P10	126	26.37	32.16	+5.79	0.383	0.575	+0.192
P11	97	24.21	31.25	+7.04	0.385	0.622	+0.237
P12	115	27.42	34.61	+7.19	0.418	0.619	+0.201
P13	115	26.20	33.26	+7.05	0.417	0.579	+0.162
P14	87	31.33	37.15	+5.81	0.535	0.635	+0.101
P15	81	30.55	34.97	+4.42	0.502	0.607	+0.105
P16	102	30.52	36.18	+5.65	0.491	0.601	+0.110
P17	71	30.91	37.30	+6.39	0.533	0.636	+0.103
P18	75	26.85	34.00	+7.15	0.451	0.621	+0.170
P19	105	26.98	34.47	+7.50	0.414	0.616	+0.201
P20	71	33.94	37.15	+3.21	0.529	0.610	+0.081
P21	110	31.57	37.25	+5.68	0.460	0.613	+0.153
P22	79	30.27	36.09	+5.82	0.463	0.590	+0.127
P23	70	29.46	34.76	+5.30	0.517	0.643	+0.127
P24	100	26.47	34.23	+7.77	0.489	0.667	+0.178
P25	135	26.42	31.31	+4.88	0.404	0.602	+0.198
All	**2384**	**29.71**	**35.68**	**+5.97**	**0.439**	**0.563**	**+0.124**

cases. The largest drop observed was −17.1 dB. Similarly, 66 slices (0.4%) showed a decrease in SSIM.

Although these cases are relatively few, they are important. They suggest that the model can occasionally introduce errors or distort important structures, which could be risky in a clinical setting. This highlights the need for careful evaluation on a per image basis before relying on GAN based enhancement methods.

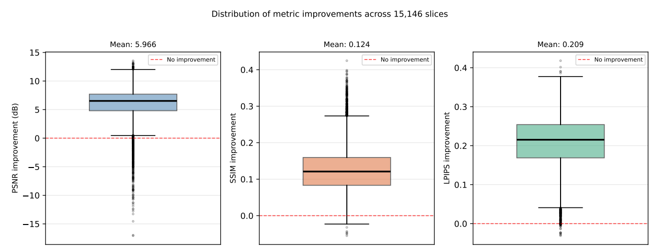

Figure 5. Box plots of per-slice metric improvements across all 15,146 test slices. Red dashed line marks zero improvement. The distributions are right-skewed for PSNR and SSIM, with the bulk of slices showing substantial positive gains.

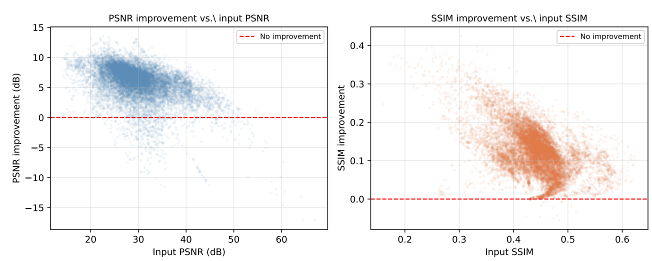

Figure 6. Per-slice PSNR improvement (left) and SSIM improvement (right) as a function of input metric value, across all 15,146 test slices. Red dashed line marks zero improvement.

3.5 Relationship Between Input Quality and Enhancement Gain

Figure 6 shows per-slice PSNR and SSIM improvement as a function of input metric value. A negative correlation is evident between input PSNR and improvement magnitude. Slices with lower input PSNR (higher noise) tend to benefit more from enhancement, while slices with already-high input PSNR cluster near zero improvement or occasionally show degradation. This pattern is consistent with the hypothesis that the generator applies stronger corrections to severely degraded inputs, but this also suggests that well-preserved slices are at greater risk of unnecessary modification. The clustering of failure cases at higher input PSNR values (above approximately 35 dB) indicates that enhancement of near-full-quality inputs can introduce unnecessary artifacts.

4. DISCUSSION

4.1 Interpretation of Results

The results demonstrate that CycleGAN consistently improves low-dose PET image quality across all six evaluated metrics and all 25 test patients. A mean PSNR gain of $+5.97$ dB, combined with a 74.6% reduction in LPIPS perceptual distance and a $2.3\times$ increase in VIFp, indicates that the generator successfully recovers a substantial portion of the structural and textural information lost through dose reduction. The consistency of improvement across patients with widely varying baseline input quality suggests that the model has learned a general noise-to-signal mapping rather than overfitting to specific anatomical patterns in the training set.

The results show a negative relationship between input quality and improvement. Lower quality inputs tend to benefit more from enhancement, which is clinically relevant since dose reduction is most important for these cases. This alignment between where the model helps most and where help is most needed is an encouraging signal for the practical utility of the approach.

However, the 3.4% rate of negative PSNR improvement across slices demands careful attention. While this fraction is small in aggregate, it is not negligible in a clinical context: 521 slices across 25 patients experienced quality degradation rather than enhancement, and the worst-case degradation of -17.1 dB represents a severe distortion that could render a slice diagnostically misleading. Examination of the scatter plots in Figure 6 reveals that failure cases are concentrated among slices with already-high input PSNR, suggesting that the generator over-processes inputs that are already close to the full-dose reference. This is a known failure mode of generative models: when the input lies near the boundary of the target distribution, the generator may apply unnecessary transformations that introduce rather than remove artifacts.

The high variance of NMSE improvements (SD $= 0.396$, nearly as large as the mean improvement of 0.277) further reflects instability in the model's reconstruction quality across slices. Unlike PSNR and SSIM, NMSE is sensitive to global intensity scaling errors and can be disproportionately affected by a small number of high-error regions.

4.2 Clinical Safety Considerations

The risk of hallucination in generative PET enhancement is qualitatively different from noise in the original low-dose image. Noise in low-dose PET is stochastic and statistically well-characterized. A hallucinated lesion or a suppressed genuine uptake focus, by contrast, is a systematic error that may be indistinguishable from true signal in the enhanced image. This distinction is especially consequential in oncology, where a false-positive lesion could trigger unnecessary biopsy or escalation of therapy, and a false-negative could delay treatment of active disease.

The evaluated metrics are all reference-based image quality measures. They can quantify how closely the enhanced image matches the full-dose reference on average, but they cannot directly identify which specific structures in a given enhanced image are hallucinated and which are faithfully recovered. A practical clinical validation pathway would additionally require lesion-level analysis: detecting all lesions in a set of full-dose reference images, then verifying their presence, location, size, and SUV in the corresponding enhanced outputs. Such an analysis is beyond the scope of the present work but represents an essential step before any deployment consideration.

4.3 Limitations

Several limitations constrain the conclusions that can be drawn from this study. First, the evaluation is restricted to a single dose reduction factor; the UDPET dataset provides up to $100\times$ reduction variants, and it is likely that performance degrades at more extreme dose reductions where the signal-to-noise ratio of the input is more severely compromised. Second, all slices in this experiment are coronal reformat orientations; it is not known whether the model generalizes to axial or sagittal planes without retraining. Third, training was performed with a batch size of 1 due to GPU memory constraints at 640×320 resolution, which may have introduced training instability relative to larger-batch regimes. Finally, no radiologist visual grading was conducted, leaving open the question of whether the metric improvements translate to improvements in diagnostic confidence as assessed by clinical experts.

5. CONCLUSION AND FUTURE DIRECTIONS

This project explores is CycleGAN improves low-dose PET scans without using paired data. A preprocessing pipeline converts raw DICOM scans into a consistent format, and the model translates low-dose images to full quality images. The results show improvements across all six metrics with substantial gains in PSNR, SSIM, and perceptual similarity. Most slices benefited from the enhancement, but a small percentage showed worse quality after processing. This highlights an important limitation. While the model can improve image appearance, it does not always behave reliably and can sometimes distort anatomical structures.

For future work, training a model that works across different dose levels would make the approach more flexible in practice. Adding constraints that preserve quantitative values such as SUV could help ensure that the output remains clinically meaningful. It is also important to develop ways to detect when the model is likely to fail, so that poor quality outputs can be avoided. Exploring newer architectures and extending the approach to full 3D volumes may further improve results. Finally, proper validation with radiologists and testing across different datasets will be necessary before such methods can be considered for real clinical use.

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